OisteBio GmbH
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Conversion Log

Production Conversion Log

May 2025 · Girardot, Colombia

Submission reference	RTFO-310825
Reporting period	May 2025 (2025-05)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,769 kg/L
Density — PLUS-grade	0,868 kg/L
Production days observed	26 of 31 (5 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

Paolo Ughetti
Managing Director

Date: ____________________

Executive Summary

Per-day kilogram-to-litre conversion log for **May 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	3 119 806,000 kg
Total EU-grade production	1 010 817,144 kg / 1 314 456,624 L
Total PLUS-grade production	1 209 153,612 kg / 1 393 034,115 L
Liquid-product yield (EU + PLUS) over input	71,16 %
Grand total volume (EU + PLUS)	**2 707 490,739 L**

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **393 677,918 kg**, steel scrap **264 326,604 kg**, water condensate **47 803,747 kg**, syngas **133 569,297 kg**, recorded losses **60 457,678 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — May 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,769 kg/L**, **PLUS 0,868 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-05-02	111 231,000	36 038,844	43 110,170	0,769	0,868	46 864,557	49 666,094	71,16 %	—
2025-05-03	137 511,000	44 553,564	53 295,597	0,769	0,868	57 937,014	61 400,457	71,16 %	—
2025-05-05	138 890,000	45 000,360	53 830,060	0,769	0,868	58 518,023	62 016,198	71,16 %	—
2025-05-06	133 529,000	43 263,396	51 752,280	0,769	0,868	56 259,293	59 622,442	71,16 %	—
Total	**3 119 806,000**	**1 010 817,144**	**1 209 153,612**	—	—	**1 314 456,624**	**1 393 034,115**	**71,16 %**	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-05-07	122 975,000	39 843,900	47 661,831	0,769	0,868	51 812,614	54 909,944	71,16 %	—
2025-05-08	126 517,000	40 991,508	49 034,616	0,769	0,868	53 304,952	56 491,493	71,16 %	—
2025-05-09	131 179,000	42 501,996	50 841,482	0,769	0,868	55 269,176	58 573,136	71,16 %	—
2025-05-10	94 582,000	30 644,568	36 657,461	0,769	0,868	39 849,893	42 232,098	71,16 %	—
2025-05-12	116 152,000	37 633,248	45 017,418	0,769	0,868	48 937,904	51 863,385	71,16 %	—
2025-05-13	128 032,000	41 482,368	49 621,789	0,769	0,868	53 943,261	57 167,960	71,16 %	—
2025-05-14	112 748,000	36 530,352	43 698,118	0,769	0,868	47 503,709	50 343,454	71,16 %	—
2025-05-15	113 137,000	36 656,388	43 848,885	0,769	0,868	47 667,605	50 517,149	71,16 %	—
2025-05-16	127 488,000	41 306,112	49 410,949	0,769	0,868	53 714,060	56 925,056	71,16 %	—
2025-05-17	86 864,000	28 143,936	33 666,170	0,769	0,868	36 598,096	38 785,910	71,16 %	—
2025-05-19	109 282,000	35 407,368	42 354,789	0,769	0,868	46 043,391	48 795,840	71,16 %	—
2025-05-20	109 298,000	35 412,552	42 360,990	0,769	0,868	46 050,133	48 802,984	71,16 %	—
2025-05-21	127 411,000	41 281,164	49 381,106	0,769	0,868	53 681,618	56 890,675	71,16 %	—
2025-05-22	118 396,000	38 360,304	45 887,132	0,769	0,868	49 883,360	52 865,359	71,16 %	—
2025-05-23	113 724,000	36 846,576	44 076,390	0,769	0,868	47 914,923	50 779,251	71,16 %	—
2025-05-24	124 830,000	40 444,920	48 380,779	0,769	0,868	52 594,174	55 738,225	71,16 %	—
2025-05-26	131 546,000	42 620,904	50 983,722	0,769	0,868	55 423,802	58 737,007	71,16 %	—
2025-05-27	143 969,000	46 645,956	55 798,545	0,769	0,868	60 657,940	64 284,038	71,16 %	—
2025-05-28	103 299,000	33 468,876	40 035,938	0,769	0,868	43 522,596	46 124,353	71,16 %	—
2025-05-29	132 402,000	42 898,248	51 315,484	0,769	0,868	55 784,458	59 119,221	71,16 %	—
2025-05-30	103 513,000	33 538,212	40 118,878	0,769	0,868	43 612,759	46 219,906	71,16 %	—
2025-05-31	121 301,000	39 301,524	47 013,033	0,769	0,868	51 107,313	54 162,480	71,16 %	—
Total	**3 119 806,000**	**1 010 817,144**	**1 209 153,612**	**—**	**—**	**1 314 456,624**	**1 393 034,115**	**71,16 %**	

Non-production days (5): 2025-05-01, 2025-05-04, 2025-05-11, 2025-05-18, 2025-05-25. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during May 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform product_densities table (one row per (product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to May 2025 are:

- **EU-grade (DEV-P100-EU)** — ρ_{EU} = 0,769 kg/L
- **PLUS-grade (DEV-P100-PLUS)** — ρ_{PLUS} = 0,868 kg/L

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform product_densities table — initially loaded by Alembic migration 0005_seed_product_densities.py and extended with per-month effective rows by 0014_monthly_densities.py. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the daily_production table:

```
litres_eu   NUMERIC GENERATED ALWAYS AS (eu_prod_kg   / 0.769)  STORED,
litres_plus NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.868) STORED
```

These columns are introduced by Alembic migration 0007_persist_production_litres.py. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to litres_eu and litres_plus is prohibited at the schema level: any attempt to update them raises PostgreSQL error code 42601.

The **input kg** column reports the daily value of daily_production.kg_to_production — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix 000000000000). The authoritative digest is reproduced in MANIFEST.sha256 at the bundle root.

— End of Production Conversion Log —